

2.2.2 Consider a linear time-invariant system with input $x(t)$ and output $y(t)$ that is described by the differential equation

$$(D + 1)(D^2 - 1)y(t) = (D^5 - 1)x(t)$$

Furthermore, assume $y(0) = \dot{y}(0) = \ddot{y}(0) = 1$.

(a) What is the order of this system?

(b) What are the characteristic roots of this system?

(c) Determine the zero-input response $y_{\text{zir}}(t)$. Simplify your answer.

2.2.3 A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^3 + 9D)y(t) = (2D^3 + 1)x(t).$$

(a) What is the characteristic equation of this system?

(b) What are the characteristic modes of this system?

(c) Assuming $y_{\text{zir}}(0) = 4$, $\dot{y}_{\text{zir}}(0) = -18$, and $\ddot{y}_{\text{zir}}(0) = 0$, determine this system's zero-input response $y_{\text{zir}}(t)$. Simplify $y_{\text{zir}}(t)$ to include only real terms (i.e., no j 's should appear in your answer).

2.2.11 A system is described by a constant-coefficient linear differential equation and has zero-input response given by $y_0(t) = 2e^{-t} + 3$.

(a) Is it possible for the system's characteristic equation to be $\lambda + 1 = 0$? Justify your answer.

(b) Is it possible for the system's characteristic equation to be $\sqrt{3}(\lambda^2 + \lambda) = 0$? Justify your answer.

(c) Is it possible for the system's characteristic equation to be $\lambda(\lambda + 1)^2 = 0$? Justify your answer.

2.3.5 Find the unit impulse response of an LTIC system specified by the equation specified by the equation

$$(D^2 + 6D + 9)y(t) = (2D + 9)x(t).$$

2.3.7 A causal LTIC system with input $x(t)$ and output $y(t)$ is described by the constant coefficient integral equation

$$y(t) + \int 3y(t)dt + \int \int 2y(t)dt = \int \int x(t)dt - \int \int \int x(t)dt.$$

(a) Express this system as a constant coefficient linear differential equation in standard operator form.

(b) Determine the characteristic modes of this system.

(c) Determine the impulse response $h(t)$ of this system.

2.3.4 Find the unit impulse response for the first-order allpass filter specified by the equation

$$(D + 1)y(t) = -(D - 1)x(t)$$

2.4.13 Using direct integration, find $e^{-at}u(t) * e^{-bt}u(t)$.

2.4.16 The unit impulse response of an LTIC system is

$$h(t) = e^{-t}u(t)$$

Find this system's (zero-state) response $y(t)$ if the input $x(t)$ is

a) $u(t)$

b) $e^{-t}u(t)$

c) $e^{-2t}u(t)$

d) $\sin(3t)u(t)$

Use the convolution table (Table 2.1) to find your answers.

2.4.21 A first order allpass filter impulse response is given by

$$h(t) = -\delta(t) + 2e^{-t}u(t)$$

a) Find the zero-state response of this filter for the input $e^t u(-t)$.

b) Sketch the input and the corresponding zero-state response.

- 2.4.23** The zero-state response of an LTIC system to an input $x(t) = 2e^{-2t}u(t)$ is $y(t) = [4e^{-2t} + 6e^{-3t}]u(t)$. Find the impulse response of the system. [*Hint:* We have not yet developed a method of finding $h(t)$ from the knowledge of the input and the corresponding output. Knowing the form of $x(t)$ and $y(t)$, you will have to make the best guess of the general form of $h(t)$.]